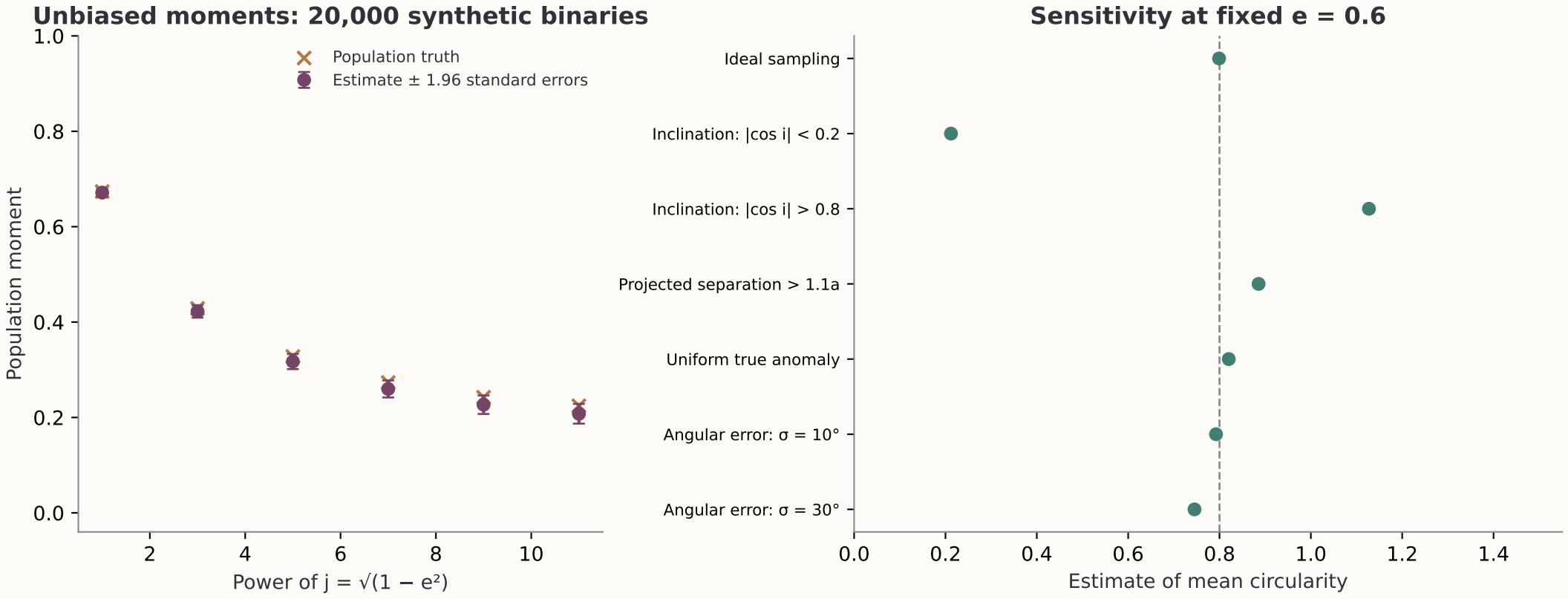
Unbiased moments: 20,000 synthetic binaries
Population truth
Estimate ± 1.96 standard errors
Population moment
1.0
0.8
0.6
0.4
0.2
0.0
2
4
6
8
10
Power of j = √(1 − e²)
Sensitivity at fixed e = 0.6
Ideal sampling
Inclination: |cos i| < 0.2
Inclination: |cos i| > 0.8
Projected separation > 1.1a
Uniform true anomaly
Angular error: σ = 10°
Angular error: σ = 30°
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
Estimate of mean circularity